DEBBIE: 161,721 low-surface-brightness galaxy candidates in DESI Legacy Imaging Surveys DR10, with a selection function measured against the catalogued galaxy population

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Abstract

The identification of low-surface-brightness (LSB) galaxies in wide-field imaging is commonly approached with convolutional networks operating on pixels. We find that the choice of representation matters more than the choice of model. Describing each source by its light profile in concentric annuli (24 radial bins, 32 position-angle cells, in g, r, and z), reduced to per-annulus statistics and low-order azimuthal Fourier amplitudes, and classifying these features with gradient-boosted trees yields AUC 0.9810 (95% CI 0.9765–0.9852) on the DeepShadows benchmark labels. Within the same pipeline and with the identical classifier, six conventional catalogue features reach 0.9098, so the difference is attributable to the representation. We make no architecture claim: the frozen artefact does not record which feature set the recurrent model consumed, and the fold-to-fold spread of a fixed model exceeds any model-versus-model difference we can measure. For context, the published DeepShadows convolutional network reports 0.9740 (published range 0.9700–0.9740) at the same sky positions and the same visually-confirmed hard-negative class, from three times our training sample; this is a comparison of complete pipelines, not of models.

Separately, we construct a catalogue from the DR10 south sweep catalogues over the complete $|b| \geq 30^\circ$ footprint (approximately 12,900 deg^2) with an explicit cut-based selection. The classifier orders the verification queue but does not define membership. We screen the result against 1.1×10^8 catalogued objects, followed by per-object queries to NED, SIMBAD, and the SMUDGes catalogue. The result is DEBBIE, a catalogue of 161,721 galaxy candidates with no prior identification as extragalactic objects in the catalogues and services queried (HyperLEDA, SDSS DR12, GLADE+, 2MASS XSC, SMUDGes, NED, SIMBAD, DESI DR1). They are not previously undetected: 99.3% carry a prior designation from an automated all-sky survey, most often WISE. What is new is the classification, not the detection.

The selection function is measured on real catalogued galaxies rather than synthetic injections. Inside the targeted window ($r_{\rm eff} = 3$–$15''$, $\mu_{\rm eff} = 24$–26.5) the photometric cuts retain 56–89% of detected-and-resolved galaxies across the 17 populated cells, and 70–89% in 15 of them, and the selection collapses above $r_{\rm eff} \approx 15''$. Three conditionals are essential and are quantified in Section 5: the figure is conditional on detection in the extract (14.2% of catalogued galaxies in the footprint, dominated by the extract's $r_{\rm eff} \geq 2''$ cut), and on lying inside the window (across all detected catalogued galaxies, most of which lie outside the 24–26.5 mag arcsec^{-2} window this catalogue targets, retention is 10.4%); and it measures the photometric cuts alone, excluding the fragment and clump filters and a size-dependent verification-completeness term of 27%. It is not a survey completeness.

Redshifts are available for 99.74% of the catalogue and all of them are photometric: removing the DESI-identified entries removed every spectroscopic redshift with them, since those spectra were DESI's own. The median object lies at $z = 0.29$ with $r_{\rm eff} \approx 16$ kpc. Most entries are therefore ordinary galaxies whose surface brightness is lowered by cosmological dimming (1.12 mag arcsec^{-2} at the median redshift) rather than intrinsically diffuse systems; 2,591 objects at $z < 0.05$ have dwarf sizes and masses. The catalogue, the code, and the 3.2 GB distilled source table from which any alternative selection can be recomputed are released in full, together with a separately flagged low-latitude extension of 78,713 candidates at $12^\circ < |b| < 30^\circ$ that inherits a galactic-cirrus caveat and is excluded from every statistic in this paper.

1 Introduction

Low-surface-brightness galaxies are under-represented in every major galaxy catalogue, and the deficit traces the selection functions of the surveys the catalogues are built from (Zaritsky et al. 2019). Recent work has treated their identification as a deep-learning problem: Tanoglidis et al. (2021) trained convolutional networks ("DeepShadows") to separate LSB galaxies from imaging artifacts in the Dark Energy Survey, establishing a benchmark on visually-labelled data.

Convolutional architectures are motivated by the absence of a known good feature basis. For galaxies, however, a strong prior exists: their light is approximately elliptically symmetric, and departures from that symmetry are themselves informative. A representation that bins light in concentric annuli encodes the symmetry directly instead of asking a network to learn it from data. This paper tests whether that representation, with an off-the-shelf classifier, matches the pixel-based deep model. We then construct a survey-scale catalogue with an explicit cut-based selection; the classifier is used operationally only to order candidates for per-object verification, and because every candidate is verified, no classifier score enters the catalogue definition.

The result of the application is DEBBIE (Deep Extragalactic Broad-Band Inference Experiment), a catalogue of 161,721 galaxy candidates in DESI Legacy Imaging Surveys DR10 (Dey et al. 2019) with no prior identification as galaxies in the catalogues and services queried. The selection function is measured on real catalogued galaxies instead of synthetic injections (Section 5.5), and the full selection can be recomputed from a released 3.2 GB source extract.

2 Data

2.1 Survey data

We use the DR10 south *sweep* catalogues: the survey's own Tractor photometry (Lang et al. 2016), comprising positions, morphological types, shape parameters, and forced photometry in g, r, z, and the WISE bands. We deliberately select from the survey's measurements instead of re-detecting sources in the imaging; Section 5.5 measures what that choice gives up in completeness.

The main catalogue footprint is DR10 south at galactic latitude $|b| \geq 30°$: all 874 sweep files in that region were processed, an occupied area of approximately 12,900 deg^2. That latitude bound is physical: at lower latitude, galactic cirrus reproduces the morphology of diffuse galaxies and foreground extinction corrupts the surface-brightness selection. These cuts and the screening of the next subsection define the catalogue on their own; the Section 4 classifier prioritised the verification order and does not affect membership. Sweeps at $12° < |b| < 30°$ were also processed; the 78,713 candidates they yield are released as a separately flagged extension, inherit a cirrus-contamination caveat, and enter none of the statistics reported here.

2.2 Benchmark data

Classifier comparisons use the DeepShadows sample (Tanoglidis et al. 2021), restricted to their objects and to their visually-confirmed hard negative class (`artifact_hard_visually_rejected`), which is the class their published AUC is evaluated against. Including the easier SVM-rejected negatives inflates the AUC of every method by approximately 0.014 and is avoided.

The comparison is asymmetric in two ways. One favours the published model: it was trained on 30,000 objects, against 10,121 available to ours. The other is unavoidable: the CNN operates on DES imagery with DES photometry, while we measure the same sky positions in DR10 imaging with our own measurement chain. The comparison is therefore between complete pipelines on shared labels, not between two classifiers on shared features; the controlled representation comparison in Section 4.2 is internal to our pipeline.

3 The radial representation

Each source is measured in 24 concentric annuli with geometrically spaced radii, each divided into 32 position-angle cells, in each of g, r, and z. Per annulus we retain the mean and the median over position angle, and the amplitudes of the first four azimuthal Fourier modes, by discrete Fourier transform over those cells [following the long-standing use of low-order Fourier terms in isophote analysis; Jedrzejewski (1987); Rix and Zaritsky (1995)], which encode lopsidedness and ellipticity without a shape fit. We keep the median as well as the mean because it is robust to a single bright neighbour occupying one wedge. Cell coverage is tracked against the count each cell would have if the frame were complete, so that brick-edge clipping registers as reduced coverage and not as a spuriously faint annulus. Profiles are extracted twice, at the catalogue position and at a flux-weighted refined centroid; both are retained, and the offset between the two centres is a feature (a large offset flags blends and fragments; Fig. 1). Classifiers use library-default hyperparameters with no per-benchmark tuning; splits, seeds, and feature definitions are in the released code.

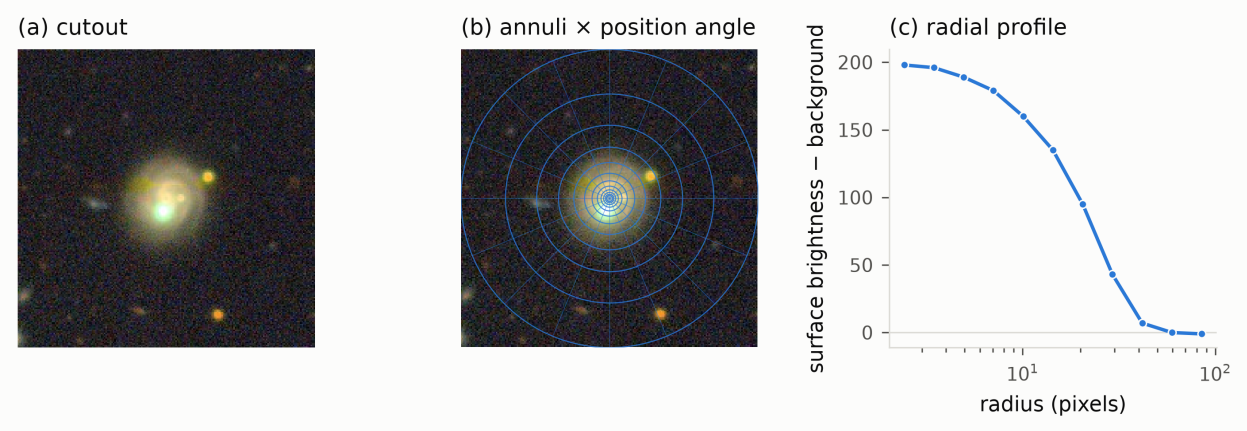


Figure 1: The representation, illustrated on a catalogue object using the released colour image (the production pipeline operates on calibrated FITS pixels): cutout (left), the annulus-by-position-angle grid (centre), and the resulting radial profile (right).

4 Classification results

4.1 Benchmark

On a balanced test set of 3,340 objects (1,670 per class), split by object and evaluated once, gradient-boosted trees on the radial representation reach AUC 0.9810 with a 2,000-resample bootstrap interval of [0.9765, 0.9852]. The published CNN reports 0.9740 [0.9700, 0.9740] on the same object and negative classes. The identical classifier on the six conventional catalogue features used in our production screening (profile monotonicity, concentration, signal-to-noise ratio, angular radius, central surface brightness, and local imaging depth) reaches 0.9098 [0.9002, 0.9198] (Fig. 2). Classifiers are XGBoost 3.4.0 and scikit-learn 1.9.0 with library defaults; splits and seeds are fixed in the released code.

4.2 Representation versus architecture

The representation is separated from the model by a controlled ablation: the same gradient-boosted classifier, the same folds, four feature sets, and nothing varying but what the classifier is shown. One fold is held out by sky position rather than at random, so that no field contributes to both sides of the split; we call it the frozen spatial fold and use it wherever a single number is quoted. On that fold, six conventional catalogue

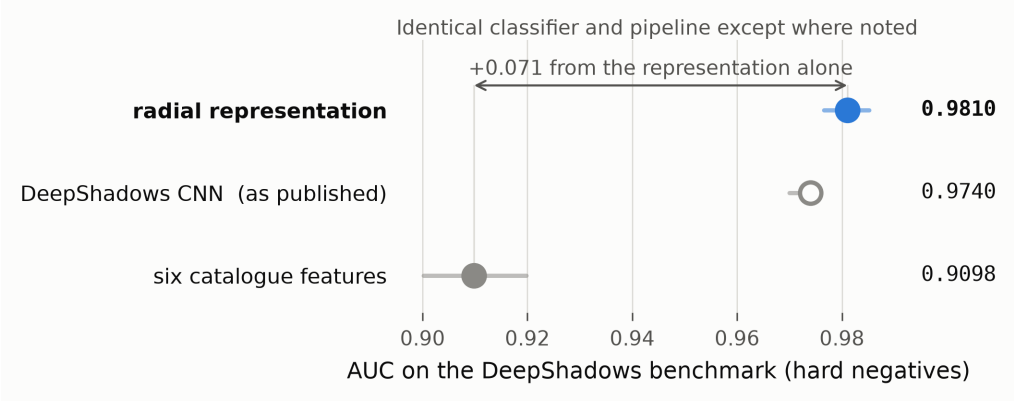


Figure 2: AUC on the DeepShadows benchmark. The gain over conventional features comes from the representation; the classifier is identical in both of our rows.

features score 0.9022, the per-annulus profile alone (144 features) 0.9645, that profile plus the six catalogue features (150) 0.9674, and the full flattened representation (1,023) 0.9775.

A single fold cannot carry this claim, because the fold-to-fold spread of a fixed model on this data is about 0.05, comparable to the effect being measured. We therefore repeated all four arms over five stratified folds, with every arm scored on every fold (Table 1). The ordering is identical on all five, and the paired gain from six catalogue features to the full representation is $+0.062 \pm 0.003$, never falling below +0.059 on any fold. The pairing is what matters: fold difficulty is common to all four arms and cancels in the difference, so the relevant uncertainty is the 0.003 spread of the paired gain and not the 0.05 spread of absolute AUCs.

Table 1: Four-arm ablation over five stratified folds, identical classifier and folds throughout.

feature set	features	mean AUC	sd
six catalogue features	6	0.9244	0.0028
per-annulus profile	144	0.9801	0.0020
profile + six catalogue features	150	0.9819	0.0019
full flattened representation	1,023	0.9869	0.0009

Two caveats bound the ablation. These folds are randomly stratified, not spatially held out, so the absolute values are optimistic relative to the frozen spatial fold above, as the two sets of numbers show, and only the paired comparison is claimed. And the ablation establishes that the representation carries information the six catalogue features do not; it does not establish that any particular architecture exploits it best. We make no architecture claim. A recurrent model was also trained on this data and scores 0.9780 on the frozen spatial fold, against 0.9775 for the trees, but the frozen artefact does not record which feature set it consumed, so that number cannot be paired with any row here.

5 The DEBBIE catalogue

5.1 Selection

From the sweep catalogues we select sources with a resolved morphological type (`EXP`, `DEV`, `SER`, `REX`; Tractor assigns `SHAPE_R` $= 0$ to PSF sources, so this excludes point sources exactly), $r_{\rm eff} \geq 3''$, mean surface brightness within the effective radius $\mu_{\rm eff}$ between 24.0 and 26.5 mag arcsec^{-2} in g, at least two exposures in both g and r, a z-band detection (a weaker, single-epoch requirement, retained because z-band epoch counts are

lower over parts of the southern footprint), and `MASKBITS` = 0. The faint boundary of the surface-brightness window matters most: above $\mu_{\rm eff} = 26.5$, Tractor increasingly fits large radii to noise fluctuations, and in 48-object contact-sheet inspections of an early pool, sources above this threshold predominantly rendered as background fluctuations (62% of that pool lay above it).

5.2 Novelty screening

Candidates are screened positionally, at 20″, against 110,190,191 catalogued objects held locally: the full HyperLEDA catalogue (Makarov et al. 2014), SDSS DR12 galaxies over the footprint (Alam et al. 2015), GLADE+ (Dálya et al. 2022), the 2MASS Extended Source Catalog (Jarrett et al. 2000), and SMUDGes (Zaritsky et al. 2019). Local screening runs at 4.5×10^5 positions s^{-1}, which permits screening every candidate. Survivors are then queried individually against NED, SIMBAD, and SMUDGes. An object whose query fails is marked unverified and re-queried; a failed check is never recorded as novelty.

Two further filters address failure modes identified during inspection. Tractor shreds extended sources, so an outer fragment can sit more than 20″ from the catalogued position. We remove fragments of large galaxies by rejecting any candidate with an entry of the 110-million-object index between 20″ and 45″. That filter also discards genuine candidates by chance coincidence: at 20,000 random footprint positions, 22.2% have an index entry in that annulus, so roughly one in five genuine candidates is removed by an unrelated neighbour. The selection function of Section 5.5 does *not* capture that loss, because it measures the photometric cuts alone. The second filter removes spurious detections clustered on defective imaging bricks, rejecting any candidate with three or more other candidates within 1.5′. One region of 6′ radius at $(\alpha, \delta) = (144.79^\circ, -2.79^\circ)$ is excluded permanently, after visual confirmation that the brick is defective.

Duplicate entries arising when Tractor splits one galaxy into several rows are merged with a matching radius of $\max(6'', r_{\rm eff})$, retaining the largest member; 2,013 entries were merged. Objects whose NED or SIMBAD queries could not be completed after retries are excluded from the catalogue entirely, so a query failure can suppress a real candidate but cannot create a spurious one.

5.3 The nature of the claim

The objects in DEBBIE carry no prior extragalactic identification of any type, under an operational definition: an object counts as identified if it appears in HyperLEDA (non-stellar), the SDSS DR12 galaxy class, GLADE+, the 2MASS XSC, or SMUDGes, or is typed as any extragalactic object (G, GPair, GTrpl, GGroup, QSO, AGN) by NED or SIMBAD (Wenger et al. 2000); prior designations are collected from NED within 0.5′ of each position and recorded per object.

One screen is absent without changing the count, and must still be disclosed: VST ATLAS DR4 (Shanks et al. 2025) classifies 98.55% of the entries it detects here as galaxies, and it was not queried. As with VII/292 below, that is a bulk automated morphological classification applied to every source in a survey footprint, not an identification of an individual object, so we do not treat it as prior identification and remove nothing for it. A reader who takes the opposite view should add it to the reasons to read this catalogue as classification rather than discovery.

A second screen was missing entirely, and it does change the count. DESI DR1 (DESI Collaboration 2025), a public spectroscopic release covering this same imaging footprint and selecting its targets from this same parent photometry, was not among the catalogues queried. Matching the catalogue against it at 2″ finds a reliable spectrum (ZWARN = 0) for 16,270 main-catalogue entries, 9.14% of the set, of which 16,014 are typed GALAXY and 116 QSO. Those objects carry a published spectroscopic extragalactic identification and are therefore not previously unidentified under our own definition. They are removed in v1.2, leaving 161,721 entries, and every number in this paper is computed on the remaining set. The released tables mark every affected row.

It is tempting to read the same match as a purity measurement, and we explicitly do not. Of the 16,270 main-set entries with a reliable DESI classification, 99.14% are extragalactic and 140 are stars. That figure

cannot be extrapolated to the catalogue. Every matched row is a deliberate DESI target, and DESI selects its targets from this same Legacy photometry using cuts (BGS, LRG, ELG) that are themselves galaxy selections. The measured quantity is therefore "of our candidates that DESI's galaxy selection also chose, what fraction are galaxies", which is largely determined by the purity of DESI's selection rather than of ours: the selection into the measured subsample is conditioned on the property being measured. The number is a fact about those 16,270 objects, each of which has a spectrum, and those objects are precisely the ones removed from the catalogue for not being novel. It says nothing about the 161,721 entries that remain. Catalogue purity is unmeasured.

A further consequence of the novelty definition cuts against the claim. cuts against the claim. 100.0% of the catalogue appears in VII/292 (Duncan 2022), the published DESI Legacy photometric-redshift catalogue from which we draw 102,274 of our own redshifts, and of the entries there carrying a star/galaxy probability, 85.4% are assigned $p_{\rm star} < 0.5$. In that narrow sense a published catalogue has already treated most of these sources as non-stellar, and we rely on it while claiming novelty. Our position is that a bulk value-added catalogue, which assigns a photometric redshift and a star/galaxy probability to every source in the parent imaging including stars and artefacts, does not constitute identification of an individual object as a galaxy: no such entry is a claim about a particular object, none is ingested by NED or SIMBAD as an extragalactic classification, and none would support a designation. Readers who take the opposite view should read this catalogue as the classification of already-detected sources instead of the discovery of new ones. We regard that reading as legitimate, and it is why the title claims absence of prior identification rather than novelty of detection. NED aggregates the identifications of most published extragalactic surveys, which is the substantive basis of the claim; we do not claim exhaustiveness over catalogues outside these services. Nearly all entries (99.3%) coincide with an automated detection in at least one all-sky survey, most commonly WISE (Wright et al. 2010), and these prior designations are recorded per object. The claim is that no queried publication or catalogue has classified the object as a galaxy. Zaritsky et al. (2019) assigned SMDG designations to roughly 7,000 objects from this same imaging on the same basis.

5.4 Physical properties

Redshifts are available for 99.74% of the catalogue, and after the DESI removal none of them are spectroscopic: the 1,640 spectroscopic redshifts the catalogue previously carried were DESI's own, and left with the objects they belonged to. This is a photometric-redshift catalogue, and every distance- dependent quantity below inherits that. 102,274 come from the Legacy Surveys photometric redshift catalogue (Zou et al. 2019) matched through the CDS X-Match service (Boch et al. 2012), and 59,020 from a gradient-boosted photometric redshift trained on our own $grzW1$ photometry for sky the public photo-z does not cover ($\sigma_{\rm NMAD} = 0.046$ against the Legacy photo-z on spatially held-out sky; a shuffled-label control degrades to 0.194). The photo-z source is recorded per object, and the two photometric estimates are never merged in any analysis that depends on redshift quality.

Distances assume flat ΛCDM with $H_0 = 70$ km s^{-1} Mpc^{-1} and $\Omega_m = 0.3$; absolute magnitudes are not k-corrected; stellar masses use the $g{-}r$ colour mass-to-light relation of Bell et al. (2003) and are factor-of-two estimates. The median catalogue object lies at $z = 0.294$ with $r_{\rm eff} = 16$ kpc, $M_g = -19.95$, and $\log(M_\star/M_\odot) = 10.52$. These are the parameters of ordinary galaxies, not dwarfs, and the explanation is cosmological surface-brightness dimming: the bolometric Tolman factor is $2.5\log_{10}(1+z)^4 = 10\log_{10}(1+z) = 1.12$ mag arcsec^{-2} at the median redshift, and for a band-limited specific intensity the corresponding factor, $7.5\log_{10}(1+z)$, is 0.84 mag arcsec^{-2}. Either value is a substantial fraction of the 2.5-mag selection window, so an intrinsically normal galaxy at the median redshift enters the window without being intrinsically diffuse. K-corrections are not applied; the argument concerns the population, not any individual object. The catalogue should be read as a census of previously uncatalogued galaxy candidates, most of them distant and dimmed, not of intrinsically diffuse systems. The exception is the nearby tail: 2,591 objects at $z < 0.05$, with median $r_{\rm eff} = 3.4$ kpc and median $\log(M_\star/M_\odot) = 8.9$, dwarf sizes and masses. Of these, 2,305 satisfy $r_{\rm eff} > 1.5$ kpc and mean $\mu_{\rm eff} > 24$ mag arcsec^{-2}, approximating the ultra-diffuse criteria of Dokkum et al. (2015), which are conventionally stated in central rather than mean surface brightness.

5.5 Selection function

The selection function is measured against real catalogued galaxies rather than synthetic injections: for each known galaxy inside the observed footprint, we ask whether the sweeps contain a resolved counterpart and whether the *photometric cuts* retain it. Real profiles, neighbours, and imaging defects enter at their true rates. The measurement covers the cuts only; the novelty screen, the fragment and clump filters, and the verification-completeness term of Section 5.5.1 are all excluded from it.

The reference population is the non-stellar entries of the same 110-million-object screening index, restricted to $|b| \geq 30^\circ$ and to sky the extract covers, matched at $3''$. Of 235,715 such galaxies, 33,595 appear in the source extract as resolved sources with $r_{\rm eff} \geq 2''$; of those, in 15 of the 17 populated cells inside the selection window the photometric cuts alone retain between 70% and 89% of them (Fig. 3b); the remaining two cells are 56% (8–10″, μ 24–25) and 63% (10–15″, μ 24–25). Cell-wise Wilson 95% intervals are 0.1–9 percentage points (half-width) for cells with at least 50 galaxies and are included in `selection_function.json` in the release. Above $r_{\rm eff} = 15''$ only one of 34 known galaxies survives. With $N = 34$ the Wilson 95% interval is 0.5–14.9%, wider than the point estimate, so this bounds the retention without measuring it. The cause is not in doubt: Tractor shreds the largest galaxies and the fragment filter then removes the pieces. DEBBIE is therefore substantially incomplete above $r_{\rm eff} \approx 15''$ and cannot constrain the abundance of large LSB systems.

5.5.1 Verification completeness

The selection function above is not the end-to-end one, and the difference is large enough to state separately. Candidates enter the catalogue only after a per-object novelty check, and that check did not run on the whole candidate pool. Of 385,295 candidates passing the cuts at $|b| \geq 30^\circ$, 281,978 were checked and 103,317 (26.8%) were not, because per-object queries are excluded once their retries are exhausted.

That loss is not uniform in size. The checked fraction falls monotonically from 74.8% at $r_{\rm eff} = 3$–$4''$ to 73.5% (4–6″), 59.6% (6–10″) and 49.4% (10–15″): the largest candidates are the least likely to have been checked, plausibly because they have more catalogued neighbours and so generate the slowest and most failure-prone queries. End-to-end, the selection fraction is therefore approximately the product of the cut retention and this term: near 10–15″ roughly $0.63 \times 0.49 \approx 0.31$, not 0.63. Correspondingly, the incompleteness above $r_{\rm eff} \approx 15''$ reported earlier is a joint statement about the fragment filter and this bias, not about the sky. 12,083 unchecked candidates with $r_{\rm eff} \geq 6''$ remain in the pool; they are released with the extract so that the omission is auditable.

Two limits bound the selection function of Section 5.5. It is completeness relative to *known* galaxies; a population absent from both published catalogues and the sweeps is invisible to it. And it cannot be decomposed into detection and selection terms, because the archived source extract retains only sources with $r_{\rm eff} \geq 2''$. The raw sweeps that would allow the decomposition are no longer held.

The catalogue is also not spatially uniform: source density falls by a factor of two to four in the band $-2^\circ < \delta < +2^\circ$ (1,405 and 1,571 objects in the two equatorial 2° bins, against 3,006–5,444 in the six neighbouring bins), where prior cataloguing in the screening pool is densest (the SDSS equatorial stripes, including Stripe 82), so a larger fraction of candidates there is screened out as already catalogued. Clustering analyses must model this.

5.6 Purity

Purity has not been measured by visual inspection. What the catalogue looks like can still be shown, and Fig. 4 shows it from both ends: eight selected entries above, eight drawn at random below. A 1,999-object sample stratified over 28 cells of the ($r_{\rm eff}$, $\mu_{\rm eff}$) plane is reserved for that, weighted so the unbiased catalogue total can be reconstructed, with an effective size of 1,776. One independent instrument does constrain purity in the meantime. VST ATLAS DR4 (Shanks et al. 2025) images part of this footprint with a different telescope, pipeline and seeing, and, unlike a spectroscopic target list, catalogues everything it detects, stars and noise included, so selection into the matched sample is on detectability rather than on the property

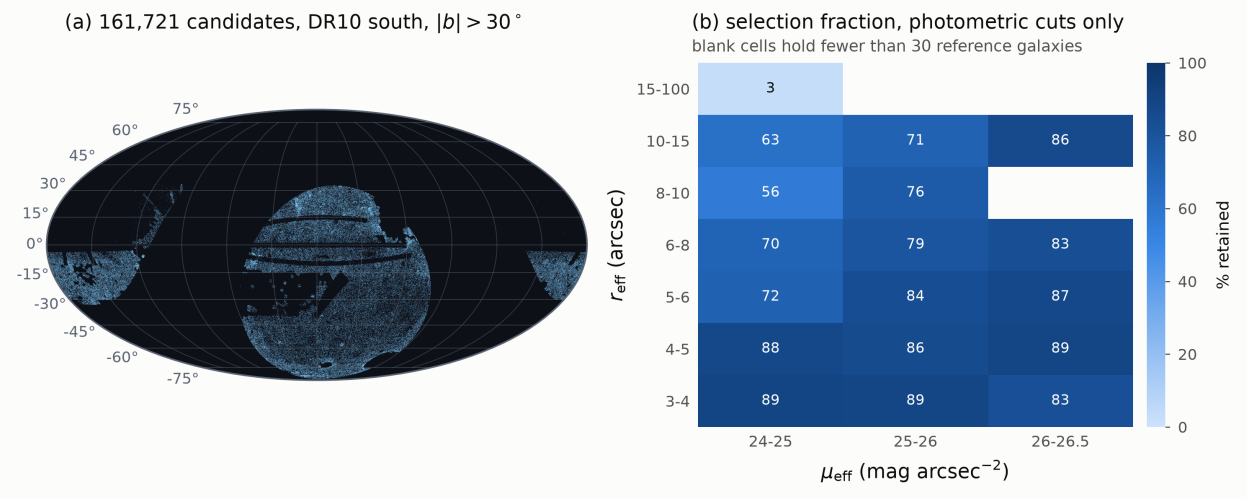


Figure 3: Left: the 161,721 catalogue objects on the sky. Right: the selection fraction for known galaxies present in the sweeps, as a function of size and surface brightness; bins with fewer than 50 galaxies are starred.

being measured. Matching 20,000 random catalogue positions at 2″ returns a counterpart for 28.0%, and of those 98.55% carry `MERGEDCLASS` galaxy or probable galaxy, 1.04% star or probable star, and 0.41% noise: an implied contamination of 1.45% (95% CI 1.14–1.76%).

Three limits stop this being the purity measurement. The matched subset is 0.21 mag brighter than the unmatched remainder, so it under-samples the faint end where contamination should concentrate. VST's classifier and ours both rest on morphological star–galaxy separation, so they are correlated methods, not independent tests. And the match reaches only the 28% of the catalogue inside the VST ATLAS footprint.

A second external check bounds contamination from below. Gaia DR3, which enters the pipeline nowhere, has a counterpart at 1.84% of 20,000 random catalogue positions, and 0.79% (95% CI 0.67–0.91%) show 5σ parallax or proper motion and are therefore certainly stars. That is a hard lower bound and a weak one: implied total magnitudes are median $g = 20.8$ and only 18.1% of the catalogue is brighter than $g = 20$, where Gaia astrometry is reliable. Within that Gaia-sensitive subset the star rate is about 4.4%. Extrapolating it, rather than the 0.79%, is the relevant comparison, and it leaves plausible contamination at several percent, which does not contradict the $p_{\rm star}$ expectation of 11.3%.

As a further interim expectation, treating the Legacy Surveys $p_{\rm star}$ as a calibrated probability gives $\sum_i p_{{\rm star},i} =$ 18,301 expected stellar contaminants (11.3%); 14,973 entries (9.3%) individually exceed $p_{\rm star} = 0.5$ and carry a flag. Entries are retained with the flag, not removed.

6 Data availability

The catalogue is released as `DEBBIE.csv` (161,721 rows) with `DEBBIE_lowb_extension.csv` (78,713 rows), version 1.2, each with an accompanying ECSV and SHA-256 checksum, containing designations, positions, sizes, surface brightnesses, colours, Sérsic indices, redshifts with per-object provenance, physical sizes, absolute magnitudes, stellar-mass estimates, and all prior automated designations. Designation numbers are stable across versions by construction and the sequence is not contiguous (merged duplicates retire their numbers). The full selection can be recomputed from the released 3.2 GB distilled source table (33.5 million resolved sources, 31 columns), which reproduces the catalogue exactly.

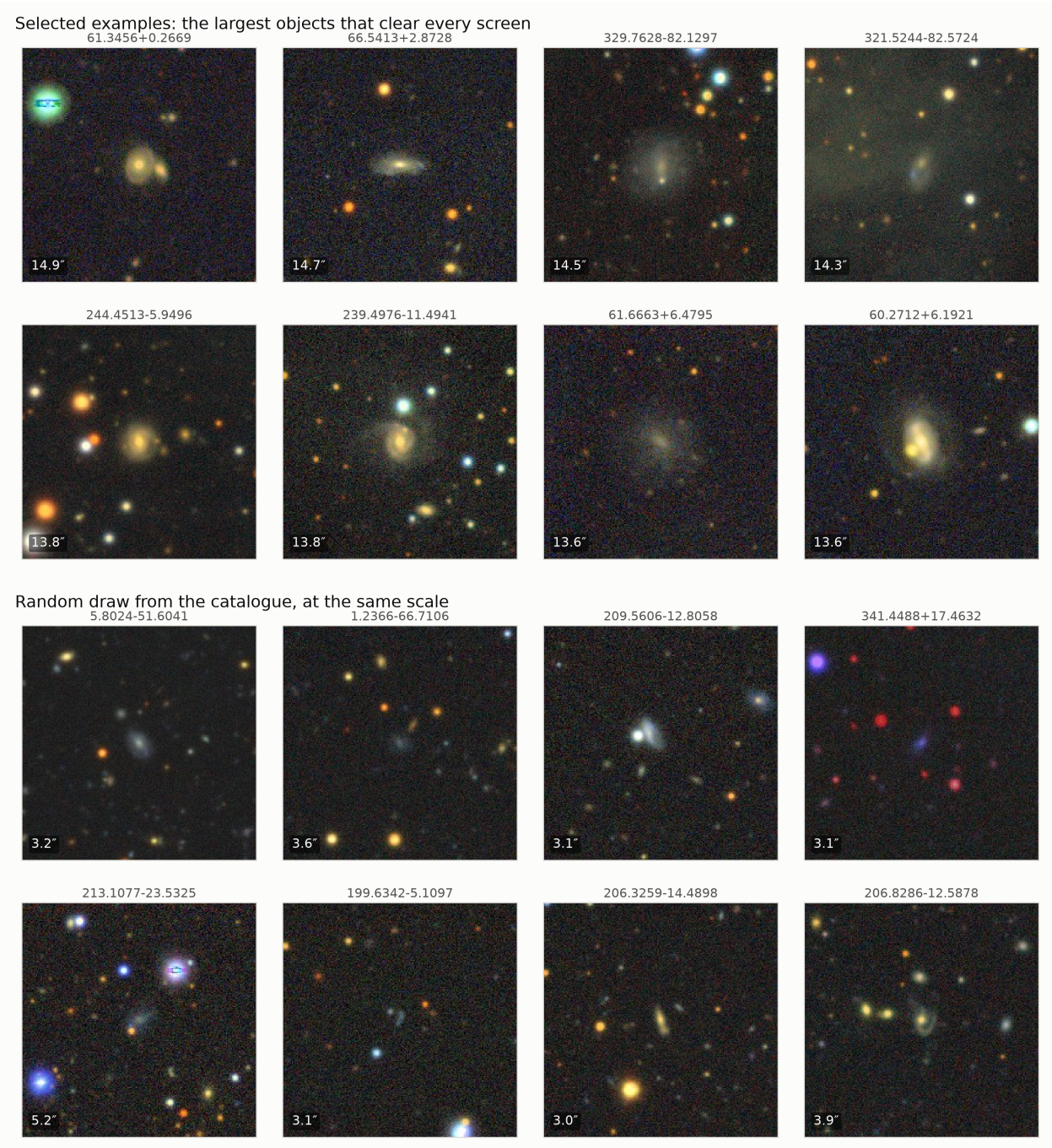


Figure 4: Catalogue objects, 75″ per panel. The upper block is eight selected examples: the largest entries clearing every screen. The lower block is eight drawn at random from the catalogue, at the same scale. The random half is fainter and more marginal than the selected half, which is what the catalogue typically contains.

7 Limitations

Photometry is Tractor's (Lang et al. 2016) and is not independently remeasured. 37% of redshifts derive from our own photometric estimator, which is trained on photometric labels; its quoted scatter is measured against the Legacy photo-z, not against spectroscopy, and these values are suitable for population statistics only. None of the redshifts are spectroscopic. Purity is unmeasured: the stratified visual sample reserved for it has not yet been inspected. The selection inherits Tractor's own detection incompleteness, collapses above $r_{\rm eff} \approx 15''$, excludes $|b| < 30^\circ$ entirely, and is spatially non-uniform where prior surveys are deepest.

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